Shweta Sharma — Data & Labels

Measuring Physician Communication at Scale

Visual companion — taxonomy maps, pipeline & label architecture, measurement quality metrics, and the NER de-identification design.

ILLUSTRATIVE VALUES All F1, κ, ECE and Brier values below are plausible placeholders — no real results exist in the project documents yet. Paste your real numbers and they will be swapped in.

~985	11	16 → 12	5 → 2	~50/wk
conversations labeled	active constructs (6 CCG + 5 NURSE)	skills identified → AI-detectable	frameworks reviewed → implemented	human throughput ceiling the pipeline replaces

01 Framework Taxonomy

Five frameworks reviewed → two implemented. Sixteen nuanced skills identified → twelve text-detectable. Four dropped — a measurement-validity decision, not a failure.

TIER 1 — FRAMEWORKS EVALUATED

Calgary–Cambridge
6 phases · 70+ behaviors
✓ PRIMARY

NURSE
Empathy micro-skills
✓ PRIMARY

SHARE
Shared decision-making
✗ SCOPED OUT

SPIKES
Bad news delivery
— REFERENCE

McLaughlin 6-Step
Assessment framework
— REFERENCE

TIER 2 — 16 NUANCED MEDICAL SKILLS, 5 CATEGORIES

The Emotional Core	Relational Mechanics	Information Gathering	Cognitive Integration	The Collaborative Plan
~~Emotional Regulation~~	Relationship & Trust Building	Active Listening	Complex Clinical Reasoning	Shared Decision Making
Empathy & Compassion	Bridging Relational Gaps	Open-Ended Questioning	Eliciting & Navigating Context	Collaborative Phrasing
	~~Bedside Manner~~	~~Nonverbal Facilitation~~		Strategic Agenda Setting
	~~Behavioral Adaptability~~	Patient-Centered Interviewing		Indirect Emotional Inquiry

12 AI-DETECTABLE SKILLS

Fully covered by CCG + NURSE together — explicit verbal behaviors, detectable in text transcripts.

4 EXCLUDED — NOT TEXT-DETECTABLE

Nonverbal Facilitation · Bedside Manner · Emotional Regulation · Behavioral Adaptability — require observation, audio, or internal state.

02 The Five-Phase Data Pipeline

Each phase is a quality gate, not just a processing step. My primary contribution area.

PHASE 1

Synthetic Bootstrap

ChatNote-style data + Label Studio. Useful for rubric testing; invalid as primary corpus.

GATE: rubric operationalized

→

PHASE 2

Real Transcript Pivot

VHA 4C anchor + ACI-BENCH, OSCE, PriMock57 supplements. ~985 conversations.

GATE: construct validity

→

PHASE 3

LLM Labeling at Scale

Rubric-grounded prompts; one construct at a time; Qwen vs. Llama selected per construct.

GATE: no cross-contamination

→

PHASE 4

PHI De-Identification

NER-based masking to i2b2/UTHealth 2014 standard. HIPAA prerequisite for real data.

GATE: ethical + legal

→

PHASE 5

SLM Training

Multi-label + ordinal 1–5 scoring. Probability outputs with audit rationale text.

GATE: formative use only

THE PIVOT — SYNTHETIC → REAL

Synthetic ChatNote-style data carried distribution-shift and model-collapse risk (Shumailov et al., Nature 2024). The corpus was re-anchored on real VHA 4C encounters, with ACI-BENCH, OSCE, and PriMock57 as supplements — a construct-validity decision, not a data-availability one.

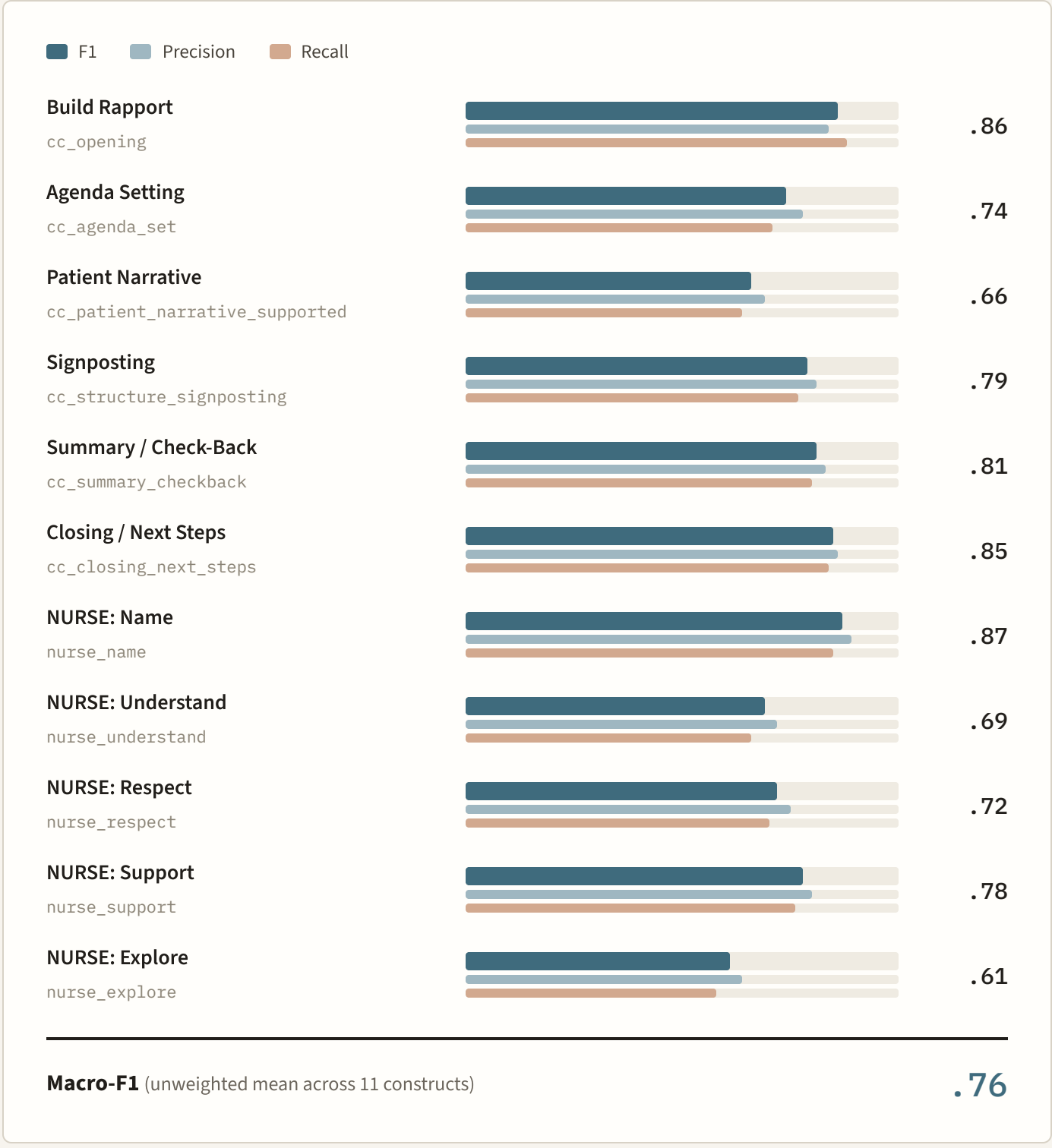
03 Label Ontology — 9 Prefix Families

Every behavioral label maps upward: transcript label → ACGME competency domain → Milestones level. Modular and composable.

`cc_` **Calgary–Cambridge** Interview structure, 6 steps	`nurse_` **NURSE** Empathy micro-skills	`sh_` **SHARE** Shared decision-making (reference)
`sp_` **SPIKES** Bad news delivery (reference)	`cr_` **Clinical Reasoning** Diagnostic parsimony, stop rules	`ml_` **McLaughlin** Teach-back, closing confirmation
`exam_` **Physical Exam** Exam quality indicators	`comp_` **ABMS/ACGME Competency** Domain scores 1–5	`ksao_` **KSAO** Knowledge, Skills, Abilities, Other

04 Per-Construct Classifier Performance

Precision / Recall / F1 per construct — aggregate accuracy is misleading for sparse, imbalanced labels. Macro-F1 is the headline metric.

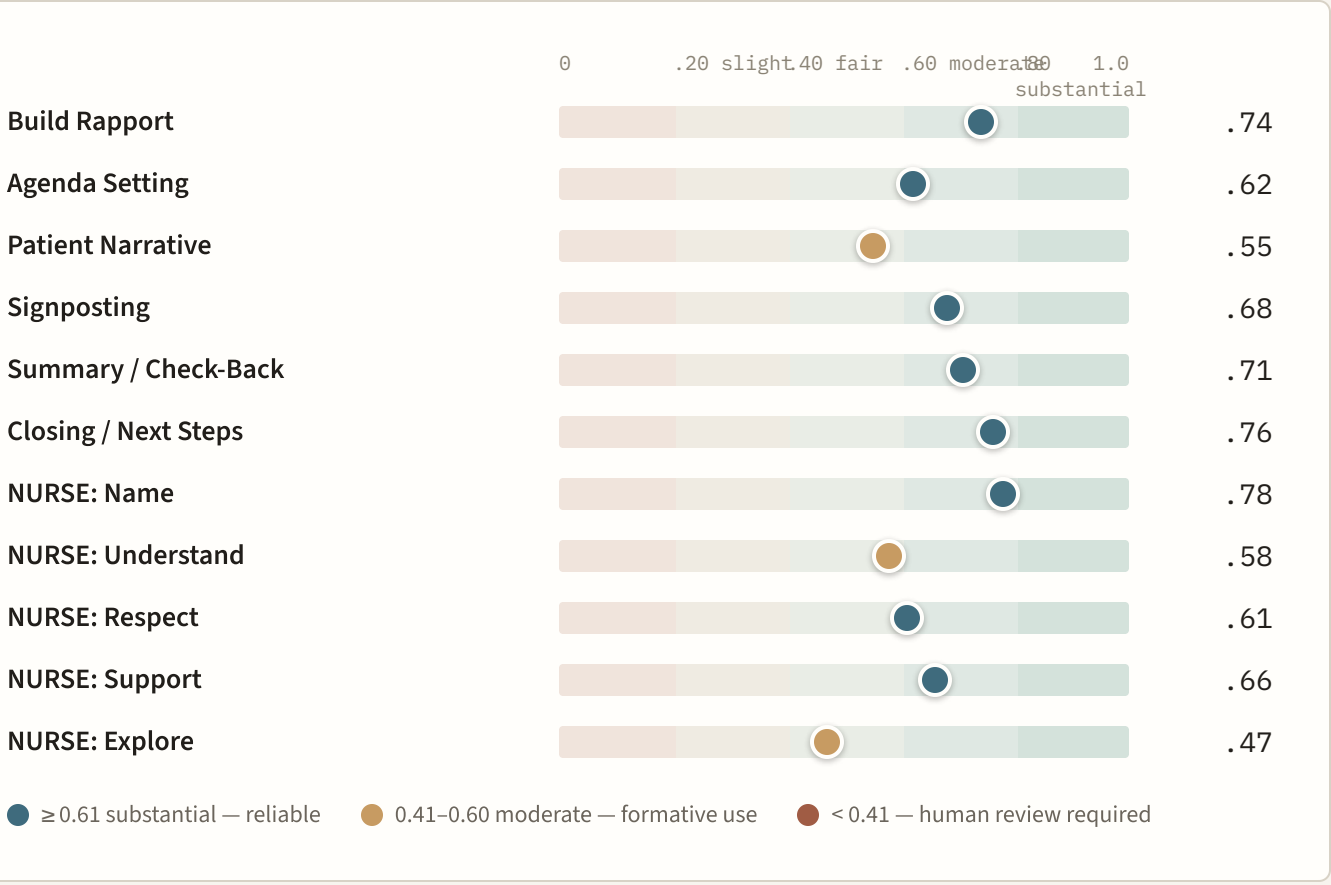


Best detected: constructs with explicit verbal markers — NURSE Naming requires specific emotion words the model finds reliably.

Hardest: constructs requiring inference — NURSE Exploring is hard to distinguish from general open-ended questioning. Itself a finding about text-only limits.

05 Inter-Rater Reliability — LLM vs. Human Graders

Cohen's κ per construct against Landis & Koch (1977) benchmark bands. Low-agreement constructs route to human review — the pipeline identifies its own uncertainty.



06 Calibration — Does the Model Know When It's Unsure?

A model that says "75% confident" should be right 75% of the time. Calibration drives the human-in-the-loop routing: auto-accept vs. review.

ECE — EXPECTED CALIBRATION ERROR

0.08 target < 0.10

Average gap between predicted confidence and actual accuracy.

BRIER SCORE

0.12 target < 0.15

Mean squared error between predicted probabilities and labels.

WHY IT MATTERS — TIERED ROUTING

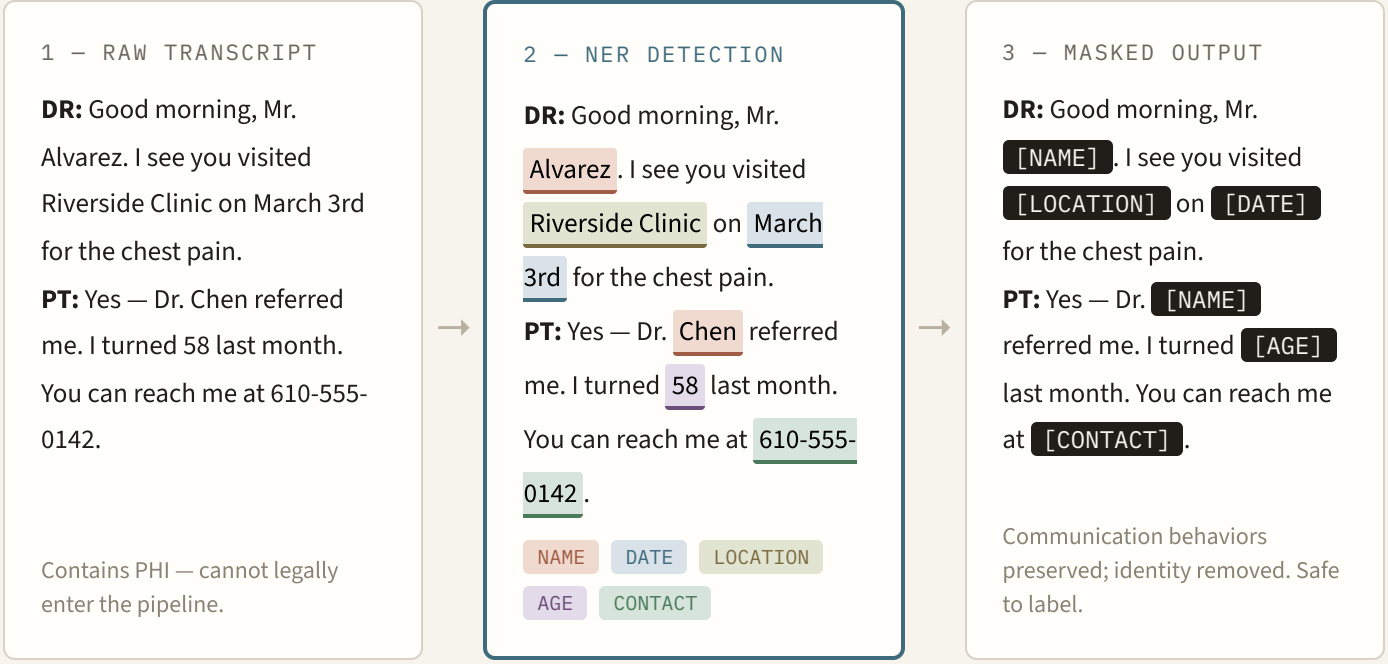
HIGH CONF → auto-accept (low-risk constructs only)

MID CONF → flag with rationale for reviewer

LOW CONF → route to expert human review

07 PHI De-Identification — How NER Works

Named Entity Recognition detects Protected Health Information spans, classifies them by type, and masks them before any transcript enters the labeling pipeline. Standard: i2b2/UTHealth 2014 de-identification corpus. *Transcript below is fabricated for illustration.*



08 NER De-Identification — Animated

The same three steps, looping: scan → classify → mask.

1 RAW INPUT | 2 SCAN ENTITIES | 3 CLASSIFY PHI | 4 MASK & RELEASE

DR: Good morning, Mr. [NAME]. I see you visited [LOCATION] on [DATE] for the chest pain.

PT: Yes — Dr. [NAME] referred me. I turned [AGE] last month. You can reach me at [CONTACT].

PHI masked with typed tokens — behaviors intact, identity removed.

Case study visuals — Shweta Sharma, ABIM Data Scientist Co-op 2025–2026. Framing follows ACGME Milestone usage guidelines: formative, developmental use only — not a validated clinical assessment tool. NER example transcript is fabricated; it contains no real patient information.